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Digital Transformation - Automation and Cost-Effective Step Towards De-Manning of Remote Facilities Leveraging Operability & Trouble Shooting Enhancement

S. Ahmed, A. Al Ketbi, K. M. Al Shehhi, H. Al Ali, F. Haider, M. Aslam, A. Al Saadi, A. Salmin, O. Alriyami, and A. Ghumara, ADNOC Onshore

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Abstract

In the oil field, assets are often located in remote areas, posing significant challenges for logistics, resource mobilization, and time-consuming troubleshooting during operational disturbances or equipment downtime. This paper presents a solution that utilizes a decision model based on Process Data obtained from the DCS and Enterprise Process Information Database. The logical model includes causes and remedies derived from FMEA analysis and operational manuals for critical equipment to facilitate fault finding and troubleshooting.

The concept involves developing a mathematical decision model based on critical operating parameters using a web-based Process system. The FMEA Decision models include respective failure causes and remedial actions, linked with the PI system to improve equipment availability and reliability. Customized dashboards were created to consolidate real-time monitoring of critical parameters, enabling proactive actions on any abnormalities. Alerts are configured to recognize deviations; when any parameter reaches an alert value, the cell color changes to yellow, and an email notification is sent with analyzed data and mathematical models, detailing causes and remedies for quick follow-up.

To realize the pilot concept, studies were conducted on failure causes, operating procedures, causes and remedial actions, and operating parameters-based troubleshooting to develop FMEA Mathematical/ Decision Models for each case. Oil-producing wells, MPFM, MSMs, multi-phase pumps, utilities, and flare systems were assessed, and all critical parameters were consolidated in customized dashboards for effective monitoring and quick analysis to avoid nuisance trips.

Benefits

The following benefits pertain to process safety, operating integrity, equipment availability, business continuity, and sustainability/emissions reduction:

- Real-time monitoring of annulus pressure ensures well integrity for over 300 wells.

- Proactive and timely operation and maintenance troubleshooting helped ensure over 30 wells' availability at greater than 95%, reducing downtime.
- Continuous production gain of 500 BOPD (25k USD) per well through real-time wells production analysis with respect to differential pressure across the choke valve.
- Ensuring uninterrupted chemical dosing system operation improved pipeline integrity by maintaining one pump operation at all times.
- Improved predictive intervention by adding alarms and analysis tools.
- Identifying all flare sources, their characteristics, and process parameters to quantify flaring volume.
- Achieving a 20% reduction in flaring, equivalent to 1050 metric tons per year, through a proactive approach.
- Ensuring operating integrity and process safety, maintaining the good health of the F&G system.
- Building team knowledge for effective troubleshooting.

Ensuring critical functions of major equipment during operational or maintenance interventions by an operator or maintenance technician requires pre-arranging necessary resources before proceeding to remote locations. This approach saves time and improves the quality of the intervention (maintenance or operations corrective actions) to prevent recurrences and maintain uninterrupted production. The successful implementation of this concept has resulted in multiple benefits for the operation of remote facilities.

Introduction

Depending on the exploration of reservoirs, oil fields consist of different satellite stations. Each station is equipped with its own facilities, including gathering manifolds, testing facilities, multi-phase pumps, pig launchers, instrument air, and nitrogen generation systems. Each station also has a relief/drain system to handle any depressurization of flowlines or relevant station facilities.

Field Background and Challenges

The purpose of surface facilities at remote stations is to gather crude from the wells and transport it to the nearest existing processing facility in Buhasa through transfer lines. Multiphase Pumps (MPP) are installed at gathering stations to transport multiphase fluid through separate transfer lines. The proposed scheme for remote fields consists of an Oil Gathering System to collect oil from multiple production wells to a Multiphase Pumping (MPP) station through a multi-port selector manifold (MSM). A Multi-Phase Flow Meter (MPFM) will be provided for well testing. The Multiphase Pump boosts the pressure of the production from wells to a level sufficient to deliver to the transfer line, approximately 21 km away. The power supply is provided through a 33 kV overhead power line from the power station.

The well fluids are gathered and collected in three remote degassing stations (RDSs). From the RDS, the fluids are transferred through piggable transfer lines to central degassing stations for further processing. All remote degassing stations are equipped with:

- Gathering manifolds
- Testing facilities
- Transfer manifolds
- Chemical injection systems
- Piggable facilities

- Flare/Ignition systems
- Instrument air systems
- Fire & Gas detection systems
- Relief and drain systems

The remote stations are manned during daylight hours and unmanned at night. They are operated using a DCS and can be controlled from the CDS control room after daylight hours using the "Remote" option via SCADA. However, only real-time data can be viewed, and no history is available.

Objective

The objective of the task was to achieve the following targets:

- Capture process data from geographically distributed fields into a centralized Real-Time Information System.
- Provide operations, technical, and management staff with a tool or system to assess, improve, and optimize.
- Provide an integration layer to other business applications the company is already using (IAOM, WIDA, SAP, PIES, PIMS, eLogs, and others).
- Visualize, report, and archive data.
- Create a user-friendly troubleshooting dashboard.
- Develop "what-if" scenarios linked to email notifications mentioning causes and remedial actions along with relevant document references.

Prerequisites

The following prerequisites were assessed to ensure the complete setup availability and its integrity:

- Data integration and availability on DCS/SCADA
- Integration with the PI system
- Integration of all critical parameters on an Excel sheet
- Developing a similar dashboard on the PI system
- Easy access for all engineers and operators
- User-friendly interfacing

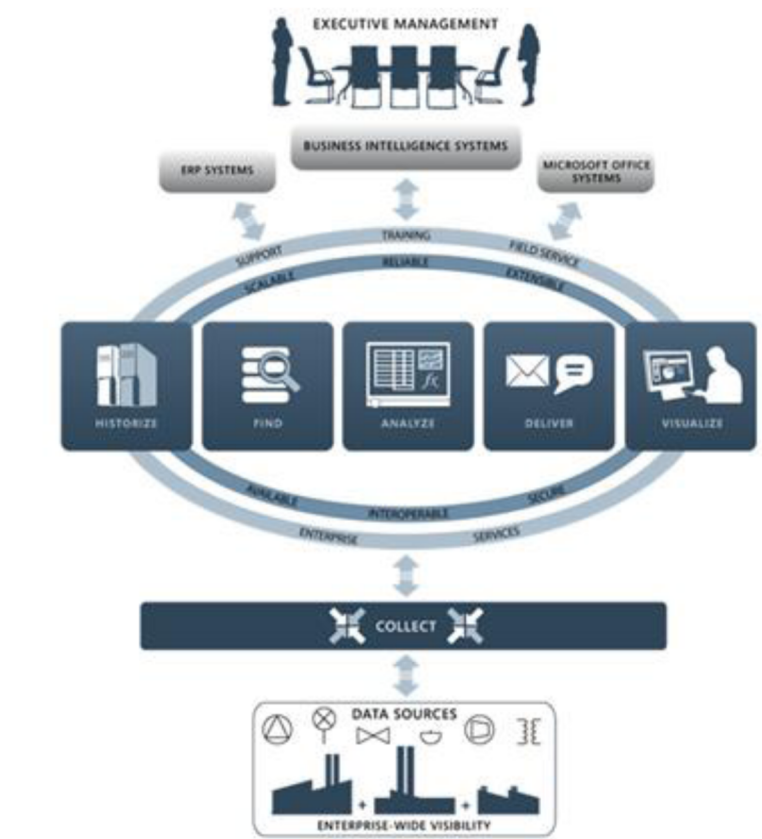
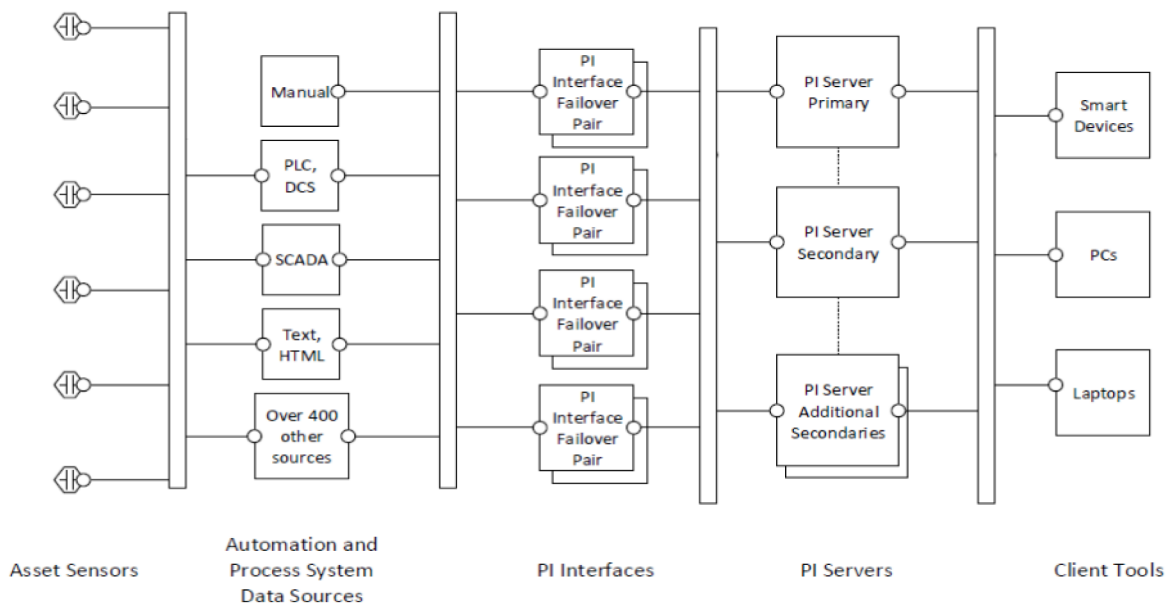


Figure 1—PI System Algorithm

The PI System (Field Data Integration System) is a suite of software products used for data collection, historicizing, finding, analyzing, delivering, and visualizing. It serves as an enterprise infrastructure for managing real-time data.

General Architecture



FDIS Client Tools

The following tools were utilized to benefit from the system and dashboard:

- PI ProcessBook: Display authoring and process monitoring
- PI DataLink: Reporting and table-based analytics in Microsoft Excel
- PI Vision: Ad hoc analysis and collaboration

Implementation

The idea was implemented in various areas of requirements:

- Stations Dashboard
- Wells & Chemical Injection System Dashboard
- Gas Lift Wells Dashboard
- Flaring Diagnostic Tool

Stations Dashboard

The remote stations' operations are vital to ensure meeting the production mandate. Ensuring integrity requires carrying out inspections and health checks of the equipment at certain frequencies. Based on the requirements and set frequencies, the facilities are designed, and shutdown periods are set to hand over equipment to maintenance. The remote station consists of the following facilities:

- Number of oil-producing wells connected through production manifolds to the suction of the pumps. Some have low, medium, and high potential concerning production, shut-in pressure, and water cut percentage.
- Each well has oil flowlines up to 3-5 km in length.
- Multi-phase pumps (MPP): Normally, one is in operation and one standby to boost the pressure up to 44 barg from 20 barg for the fluid coming from the wells to transport it to a nearby station with a landing pressure of 30 barg.
- Inlet manifold where the flowlines come from both directions of the station to connect with the MSMs and production & test headers.
- Two production/test headers
- MSM (Multi Selector Manifold): Each MSM has provisions to connect seven transfer lines from the wells, with some wells sharing lines to accommodate more wells to the station.
- Discharge header with a provision for a pig launcher.
- MPFM (Multi-Phase Flow Meter): To test the potential of individual wells in terms of oil flow rate, water flow rate, and gas volumes.

MSM U-0101		MSM U-0102		MSM U-0103		MPFM 1601		MPFM 1602		Discharge Parameters	
BQ-24/34	2.00	BQ-10	2.00	BQ-5/41	2.00	Inlet Pressure, Barg	21.20	Inlet Pressure, Barg	21.80	Discharge Header Pressure-A, Barg	40.50
BQ-35	2.00	BQ-17	2.00	BQ-43/51	2.00	Flow Meter Outlet Pressure, Barg	21.86	Flow Meter Outlet Pressure, Barg	21.82	Discharge Header Pressure-B, Barg	40.70
BQ-30	2.00	BQ-44	0.00	BQ-42	0.00	Outlet Temp, Deg C	36.99	Outlet Temp, Deg C	61.90	Discharge Header Pressure-C, Barg	40.53
BQ-07	2.00	BQ-47	2.00	BQ-14	2.00	Prover Unit Outlet PCV Output, %		Prover Unit Outlet PCV Output, %		Pressure at d/s of SDV-1301-11	40.45
BQ-19	2.00	BQ-8	2.00	BQ-09	2.00					Pressure at d/s of SDV-1301-12	40.40
BQ-26	2.00	BQ-28	2.00	BQ-29	2.00					Pig Launcher LP 1301 Pressure, Barg	3.72
BQ-27	2.00	BQ-18	2.00	BQ-16	2.00						
MPP-1				MPP-2				Closed Drain And Flare System			
Production Header-1		Pump Across DP, Bar	20.77	Production Header-2		Pump Across DP, Bar	0.13	P-6701 Discharge Pressure, Barg	4.48	Pressure From Propane Cylinders	0.13
Inlet Pressure, Barg	21.20	Pump Across DP, Bar	20.89	Inlet Pressure, Barg	21.63	Pump Across DP, Bar	0.11	P-6702 Discharge Pressure, Barg	3.23	Flow Rate From Propane Cylinders Nm3/hr	0.11
Inlet Temp, Deg C	42.94	Pump Discharge Temp, DegC	43.74	Inlet Temp, Deg C	28.73	Pump Discharge Temp, DegC	28.98	P-6703 Discharge Pressure, Barg	40.67	Gas Flow From KOD, Nm3/hr	28.98
Suction Pressure-A Barg	21.06	Pump Discharge Temp, DegC	43.91	Suction Pressure-A Barg	22.60	Pump Discharge Temp, DegC	29.44	P-6703 Discharge Pressure, Barg	0.02	Gas Flow From KOD Totalizer Nm3/hr	29.44
Suction Pressure-B Barg	21.19	Pump Discharge Pressure, Barg	40.52	Suction Pressure-B Barg	22.38	Pump Discharge Pressure, Barg	21.25	Discharge Common Header Pressure Barg	40.72	MPP Auto Foam System	
Suction Strainer-1 dp, Bar	1.23	Pump Discharge Pressure, Barg	40.65	Suction Strainer-1 dp, Bar	0.01	Pump Discharge Pressure, Barg	21.28	Closed Drain KOD Level-1, %	Configure	Foam Discharge Pressure, Barg	21.28
Suction Strainer-2 dp, Bar	1.18	Speed Setpoint RPM	704.00	Suction Strainer-2 dp, Bar	0.00	Speed Setpoint RPM	300.00	Closed Drain KOD Level-2, %	41.96	Foam Discharge Pressure, Barg	300.00
Suction Temp, DegC	47.49	Actual Speed RPM	704.00	Suction Temp, DegC	29.75	Actual Speed RPM	0.00	Closed Drain KOD Level-3, %	29.53	WI Clusters	
Pressure after Strainer, Barg	19.69	Actual Current Amps	154.53	Pressure after Strainer, Barg	21.25	Actual Current Amps	0.10	Closed Drain KOD Level-Average, %	35.74	Surface Pump Cluster # 1 Flow Rate, BPD	
Pressure after Strainer, Barg	19.58	Control Suction Pressure SP, Barg	20.00	Pressure after Strainer, Barg	21.37	Control Suction Pressure SP, Barg	20.00	Closed Drain KOD Pressure, Barg	0.00	Surface Pump Cluster # 2 Flow Rate, BPD	
Running Hours		Control Suction Pressure PV, Barg	19.58	Running Hours		Control Suction Pressure PV, Barg	21.46			Total Water Inj. Flow Rate, BPD	

Figure 2—Station Diagnostic Dashboard

Flaring Diagnostic Tool

Similarly, the flaring diagnostic tool is established to identify the direct root cause of flaring, helping in pinpointing and taking timely action to avoid unnecessary flaring. The following aspects were considered:

- Flaring is essential for safely releasing process fluids during upsets and trips.
- Minimizing flaring conserves energy and protects the environment.
- The sustainability pillar aims to achieve zero flaring.
- The flare network is complex, with several sources. The first step in flare reduction is identifying and quantifying the sources instantly, providing the status of all flare sources, and initiating alerts for undesirable flaring.

The following mechanisms were developed to benefit from the work done:

- Creating a traffic light system in PI ProcessBook for initiating alerts.
- Monitoring associated flaring sources (PCVs/PSVs) from a single page and taking prompt action to avoid unnecessary flaring. PCV opening outputs are configured in the graphics/sheet to indicate when the valve opens. If any PCV opens, the cell color changes to red, allowing proactive action to avoid flaring.
- Swift identification of flaring sources and facilitating corrective action.
- Using the tool's results in regular site operation meetings for visual management boards.
- Developing visual management boards (highlighting daily/weekly/monthly flaring) for visibility and effective communication. These boards are fixed in stations and displayed on visual screens, assisting in close follow-up and maintaining flaring within limits. The status of actions can be highlighted/updated if related to flare.

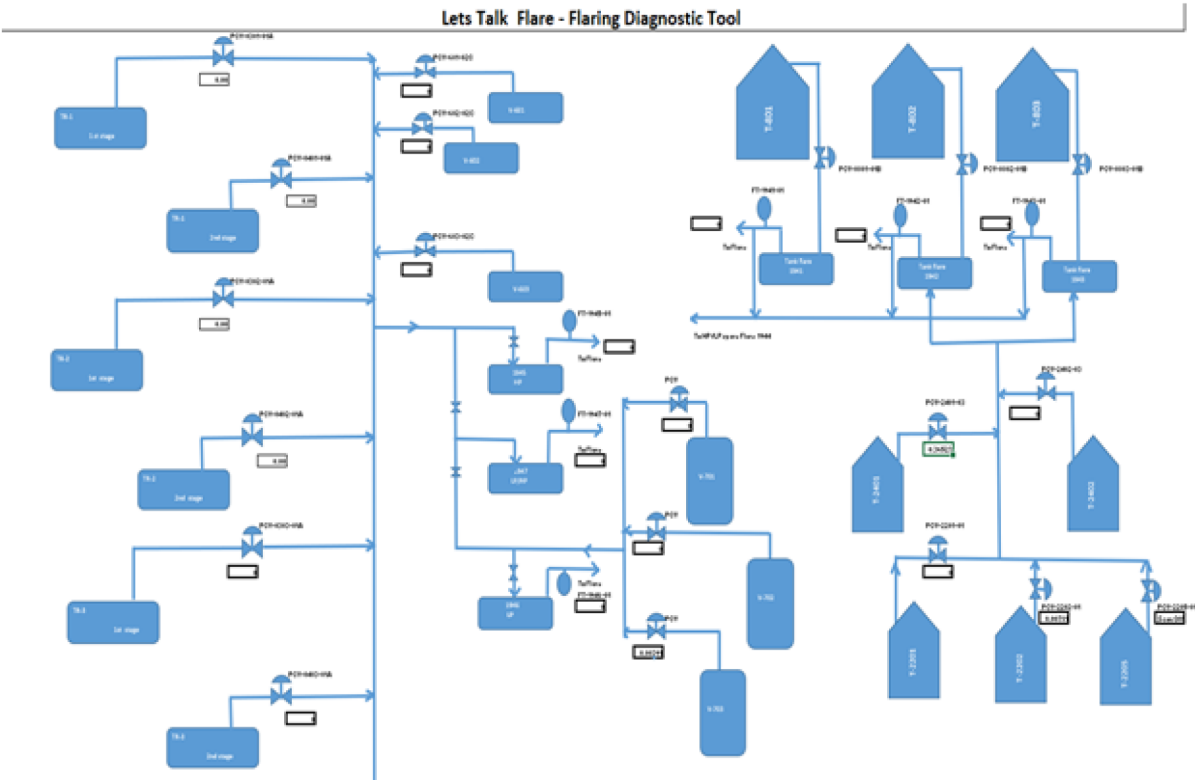


Figure 3—Flaring Diagnostic Dashboard

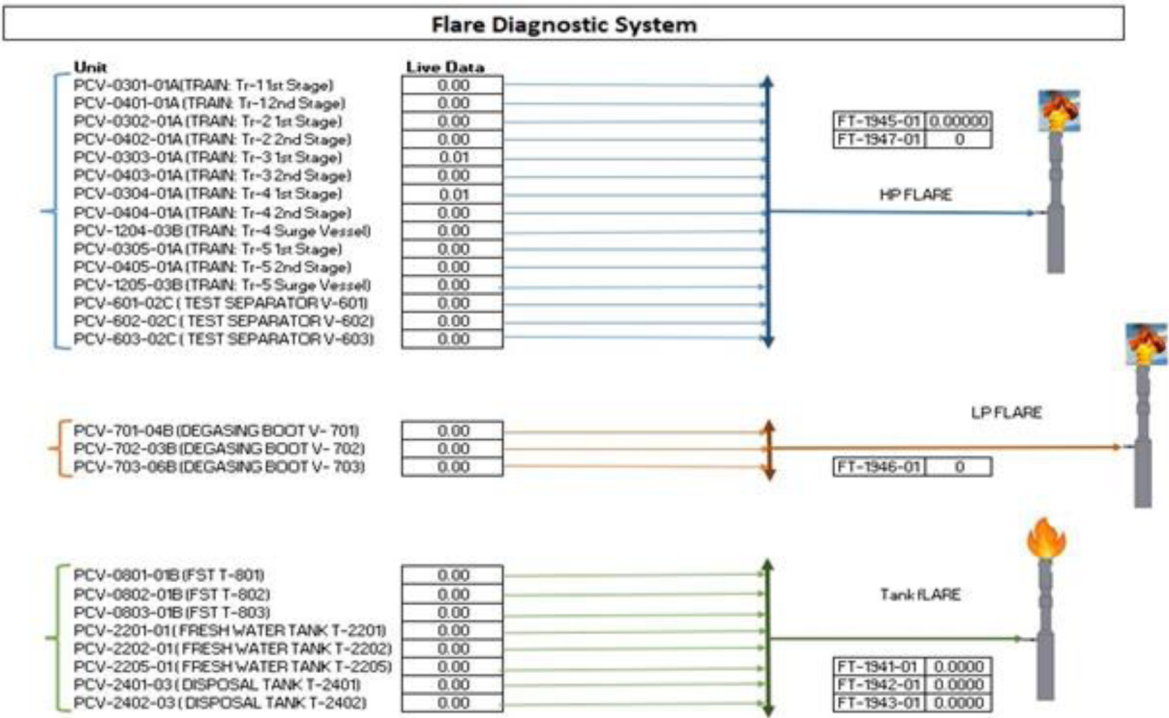


Figure 4—Flaring Diagnostic Tool

A similar analysis was carried out for gas lift wells, and configuration and integration were deployed for gas lift wells, leveraging real-time data availability on the PI system. The same was updated with the operating status confirmed from the respective data logging tool.



The following interfaces are located in the CCR:

- DCS/ESD HMI for CDS plant
- SCADA HMI
- Engineering workstations, printers, video copiers, and servers
- Fire & Gas mimic panels

- Package unit PLCs
- MOL Pumps unit control panels
- Power station
- Gasco compressor substation
- New and old electrical substations
- Partial valve stroking/testing system

- **DCS (Distributed Control System):** Based on the Invensys I/A series system, it implements all non-safety-related process control and sequences, providing operator interface and management information about the CDS plant. It also interfaces with third-party equipment using serial line connections (Modbus RTU protocol).

- ESD (Emergency Shutdown System): Implements safety emergency shutdown sequences and is based on the Tricon system, using the Triple Module Redundancy philosophy. It is integrated into the DCS redundant process communication LAN (Nodebus) using redundant I/A series Advanced Communication modules (ACM). The ESD system shares the DCS operator interface.
- F&G (Fire and Gas detection system): Interfaces with gas and flame detectors and visible/audible equipment to protect against fire and gas presence risks. It is based on the Tricon system, using Triple Module Redundancy. The F&G system shares the DCS operator interface.
- SCADA system: Remotely monitors PLCs and RTUs located in three remote degassing stations, 48 water supply wells, nine boost oil wells, and Huwaila-44. It is based on a Foxboro SCADA system running IU SCADA W software. The field equipment is based on the Tricon system for the RDS stations and IU SCADA RTUs for other locations. The SCADA system has its own operator interfaces (two units). It is connected to the DCS system using a redundant 100MB Ethernet connection on the information LAN of the DCS system.

SCADA

The main objectives of the SCADA system for fields located in the control and technical rooms are:

- Quick, efficient, and safe operations of the water injection system
- Water injection data gathering and monitoring
- Water injection system optimization
- Quick, efficient, and safe operations of the remote gathering station test separators
- Monitoring and control of remote station

The main functionalities of the system are:

- Redundant communication handling by means of terminal servers and communication line sharing devices
- Data acquisition (analog, RTD, digital) and data sending (digital and analog control) from/to remote RTUs by means of a dedicated optical fiber data highway
- Data checking and validation, real-time variable calculations, and data presentation in alphanumeric, bar chart, and trending formats
- Date and time synchronization by GPS device throughout all SCADA devices
- Historical data archiving and archived data presentation in various formats
- SCADA equipment diagnostic and recovery

The main equipment of SCADA includes:

- SCADA servers in backup configuration
- Two dedicated independent workstations
- Independent engineering workstation
- Dedicated printers for alarms, events, and reports
- Redundant LAN realized through two hub/switches
- Communication system using redundant terminal servers and serial sharing devices

Deployment Methodology

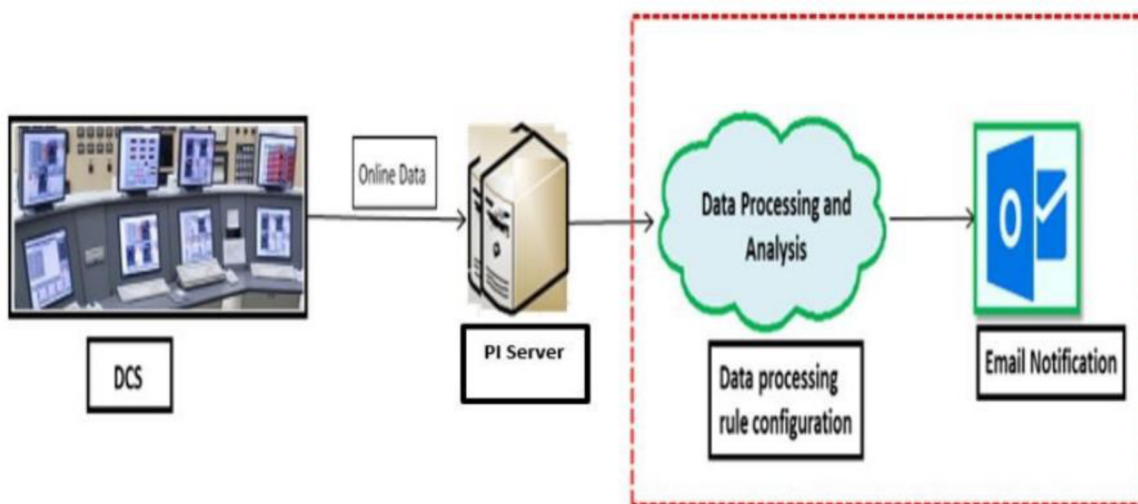


Figure 6—Logic/Flow Diagram

Email Notifications

Developing the cause and remedial action associated with any particular alarm and deviating parameters. The pattern followed for each alarm is as follows for a single tag:

- 5410PI1001_25A (MPP-1 Suction Pressure)
- Operating pressure = 20 Barg
- High Alert = 21 Barg
- High High trip = 24 Barg

Reasons for High Pressure Alert

- Increase in flow from any flowing wells
- Any new well is opened
- Running MPP tripped

Remedial Actions at High Pressure Alert Alarm

- Confirm and counter-check the pressure readings from other associated PTs (5410PI1001_25B and 5410PI1001_01)
- Check the MPP control mode from DCS, increase MPP RPM if possible (Switch from SPC to RPM Mode to increase speed manually)
- Check the suction pressure setpoint for MPP from UCP and DCP, it should be 20.0 barg (SP can be changed below 20 Bar to increase speed while in SPC Mode)
- If the suction pressure is increased due to MPP trip, ensure to isolate all flowing wells, then depressurize excessive pressure from the closed drain drum and proceed for pump startup as per SOP.

References

- BQ Operating Manual 30.99.09.1006
- BQ MPP startup procedure 30.99009.1085
- Multi-Phase Pump P-1001 P-1002 54-10-08-0106-1

Upon email notification to the individual operator for a particular tag, it is easy for the operator to understand the cause of the alert and take appropriate action on time to avoid production loss or equipment downtime. The same is developed for wells, stations, flaring, etc.

It not only assists the operator in identifying the root cause but also aids the maintenance team in troubleshooting for corrective action and reinstating the equipment into operational condition. It further enhances operability and maintainability based on the predictive model.

5410PI1001_25A (MPP-1 Suction Pressure)

- Operating pressure = 20 Barg
- High Alert = 22 Barg
- High High trip = 24 Barg

Reasons for High Pressure Alert:

- Increase in flow from any flowing wells
- Any new well is opened
- Running MPP tripped

Remedial Actions At high pressure Alert alarm:

- Confirm & counter check the pressure readings from other associated PTs (5410PI1001_25B and 5410PI1001_01)
- Check the MPP control mode from DCS, increase MPP RMP if possible
- Check the suction pressure Setpoint for MPP from UCP and DCP, it should be 20.0 barg.
- If the suction pressure is increased due to MPP trip, ensure to isolate all flowing wells, then depressurize excessive pressure from closed drain drum and proceed for pump startup as per SOP.

References:

BQ Operating Manual 30.99.09.1006

BQ MPP startup procedure 30.99009.1085

Figure 7—Email Notification sample

PT_XXXP_25 (Gas lift Well – Lift Gas Pressure)

- Normal Operating Flow range = 100-145 Barg
- Low Alert = 45 barg
- High Alert = 156 barg

Reasons/Causes for Lift Gas High Pressure Alert:

- Gas Lift MOV does not function properly.
- Gas Lift MOV stuck in Open or close conditions.
- SSSV closed
- GLV stuck
- Malfunctioning / closing of Oil Choke valve

Remedial Actions for Lift Gas High Pressure Alert:

- Check for gas side choke valve functionality from DCS or well side HMI – if erratic behavior to be attended by Maintenance team
- Check the status of SSSV – Ensure it is fully open – If not fully open, to be highlighted to PE/Wireline team for functional checks.
- If gas side high high pressure is observed, unload the well by reducing Annulus A pressure upto 50 barg and again slowly pressurize with lift gas – if HH pressure problem persists – it is to be highlighted to PE/Dev team for further investigation on GLV

Detailed Procedure for Unloading the Well

If GL pressure is high and well not taking gas and well is unable to flow, then below actions are required.

- Close manual isolation valve U/S SDV.
- Close wing valve
- Open bullhead line and de-pressurize the Annulus-A (off-load) up to 50~ 60 barg
- Close the bullhead line back to normal conditions.
- Put gas FCV on manual mode @ 52 % ~ 55% and open the SDV U/S manual valve back to normal and slowly pressurize the Annulus-A casing.
- Monitor /observe WHCIP at SWAB valve continuously, if wellhead pressure increases trend, it means GLV is working.
- If WHCIP does not increase even Annulus-A pressure (indicating on MSAS reading at HMI) reaches up to 90~95 bar. Then it may be problem on GLV or may be SC-SSSV is in close conditions.
- Check SC-SSSV needle status if normal (open positions) then close the SCSSSV from WHCP and close gas lift SDV manual valve for further troubleshooting.
- Open back SCSSSV from WHCP & record SCSSSV hydraulic oil supply up to 5000 psig

Figure 8—Email notification sample for gas lift wells

General Benefits

The following benefits are accomplished with effective utilization of this tool:

- Pinpointing abnormalities with location (tag) details. Automated email notifications with causes and remedial actions.

- Zero investment cost using available resources
- Guiding the maintenance team with probable root causes and required actions. Close follow-up on corrective maintenance requirements.
- Real-time monitoring tool for operations engineers/operators and the maintenance team.
- Single-page consolidated dashboard
- Trainees and engineers can memorize operating parameters and pinpoint abnormalities quickly. Knowledge conservation.
- Alert and trip alarms are configured to know the deviation of operating parameters. Proactive actions can be taken.
- Monitoring operating/critical parameters from a single worksheet. Quick observation of all parameters without exploring page by page. Close and effective monitoring.
- Effective follow-up on corrective maintenance requirements. Equipment availability.
- Trainees and engineers can memorize operating parameters and pinpoint abnormalities. Learning tool.
- Alert and trip alarms are configured to know the deviation of operating parameters. If any parameter is in alarm, the cell color changes to pale yellow, and if any parameter reaches the trip level, the cell color changes to red. Proactive approach.
- Engineers can advise operators to take proactive action for any abnormality/deviation in operating parameters. Proactive approach.
- If the operator is outside for some job, the operations engineer can observe and advise action to the operator to avoid any trip, preventing production loss.
- Each alarm triggers an email notification to the BQ Ops group. Operators/engineers can understand the alarm reason/cause and respective remedial actions to avoid nuisance trips. Proactive action/learning tool.

Operational Examples of Benefits Achieved

F&G System Process Safety

Observed alarms on GL wells, highlighted/raised SAP notifications, and resolved issues within two days through close follow-up. Ensured continuous F&G availability at the wellhead and production continuity from the wells.

Communication: Process Safety

Pilot wells/RFC-1 wells lost communication. Immediately highlighted the concern to the telecom team, and with close follow-up, resolved the issue. Secured real-time monitoring of wells, ensuring no alarm while approaching the wellhead as per recommendations in the respective DGS.

Production

To obtain actual production from GL and BQ, concerns were highlighted to PE that wells (e.g., BQ-26 & BQ-29 LS) were unable to flow. Observed through flowline temperature, which was in alert color due to low temperature. Anticipated production gain: 500 BOPD per well.

Chemical Dosing: Integrity Related

Ensured continuous injection of chemicals and optimized flows wherever required at the station and wellhead. Ensured CI tanks were filled on time. Ensured one pump availability and continuous operation.

Corrective Maintenance Follow-up: Equipment Availability/Business Continuity

Followed up with the IC team to resolve the BQ Nitrogen package oxygen analyzer issue, ensuring standby package availability.

Nitrogen Package Process Safety/Integrity

Package 1 inlet air heater was found in trip condition. Informed the operator to intervene to ensure good quality nitrogen for purging. Ensured pipeline integrity. Observed oxygen content on the higher side, intervened, and informed the operator to take immediate action to switch to another package to ensure uninterrupted nitrogen supply to the KOD/relief header. Another time, the nitrogen system heater was tripping frequently, and the PLC was hanging up. Troubleshooting was done as per the requisite remedial actions, ensuring uninterrupted purging of the flare header.

Sustainability/Emissions Reporting

KOD to flare vent gas flow meter FI-6701-01 showed faulty (negative reading), and the operator couldn't calculate the total accumulative gas flow through the flare (flare system continued purging with N₂). The problem was promptly identified and rectified by the maintenance team.

Uncertainties in Deployment

Some data was not interlinked with PI system however it was available on DCS and SCADA. Leveraging the data availability, the same parameters were configured on PI system and added into Excel sheet.

Server Connectivity

With Telecom support, the connectivity of real time monitoring system was ensured by deploying uninterrupted power source and loop.

Access to PI System & Excel PI Datalink

All the stakeholders were enlisted and access was provided for effective utilization.

Expertise of the crew and operators

The expertise of the involved crew evaluated on the basis of their experience. Moreover, work instruction developed for alignment between two parties in order to ensure safe and smooth operation.

Station Digitalization

The objective was utilized to define the initiatives with the execution strategy for the remote operation of facilities. This requirement aligned with company digitization objective and also to develop any short term solution to address the same.

Several discussions was held between Operation and Maintenance team to draft such road map which forms as a reference to this strategic initiative.

The agreed way forwards implemented,

- Initiatives which can be actioned by Maintenance instrument team are done immediately and will realize the benefits.

- Activities which are covered under project are reviewed with the schedule and explored possibility of short terms solution like utilizing 5G.
- The activities which are not covered in the project and requires feasibility study and Techno-economic review need to be addressed separately as a project.
- Any deviation from current operation philosophy need to be aligned with concerned approval authorities.
- Operating philosophy of these facilities need to be amended after completion of all the above initiatives.

The gap analysis was carried out for remote facilities to evaluate and develop the requirement of further digitalization and effect control to gain following perks.

- Reduction in transportation.
- Reduction in GHG emissions
- Reduction in work load of operator
- Effective monitoring and control remotely
- Manpower utilization for other upcoming projects.
- Muliskill manpower
- Integrated plan for corrective and preventive maintenance from Maintenance team.
- To align with companies' digitization objective

The aim was to make the system smart and utilize deputed manpower for other populated projects. It was recommended to interface the same with digitalization project or to be handled separately under separate project.

The main target and related scope was essentially to digitalize equipment related to remote station to gain maximum efficient operational benefits as mentioned above.

Transportation GHG emissions

With on going practice, the daily visit for remote station is being carried out for routine and non routine tasks which involves the movement of transportation. By converting the remote station operation into smart, the transportation to remote facilities will be reduced which will ultimately result in reduction of GHG emissions.

- Round trip to Remote station = 94 Km
- Operation teams drives 5,640 Km/month
- Maintenance teams drives 6,298 KM/Month
- Total km = 11,938 Km/month
- Total CO2 Emission = 3,430 KG/month

The daily visit will be reduced to 2 or 3 times in week depending on the corrective and preventive maintenance

- Round trip to Other remote Statoin = 96 Km
- Operation team drive 2,880 Km/month

- Maintenance teams drives 3,456 Km/month
- Total km = 6,336 Km/month
- Total CO₂ Emission = 1,821 KG/month

The daily visit will be reduced to 2 or 3 times in week depending on the corrective and preventive maintenance

Timely response from Operator & Maintenance

- Both stations are remotely located which possess a lot of challenges related to daily routine / non routine logistics and resource mobilization and lot of time was consumed in front end troubleshooting during any operational disturbance or equipment downtime. The smart field project implementation can make the equipment fault finding or troubleshooting process easier.
- Preserving the critical functions of the major equipment and during any operational or maintenance intervention by operator/maintenance technician, early morning before proceeding to remote location, the requisite resources can be proactively arranged which will save the time and also the quality of intervention (Maintenance / Operations corrective actions) will be executed to avoid the recurrences.

Deployment of manpower in other projects

The number of lift gas wells are increasing time by time and there is another major projects AiP5 where the facilities are being expanded, the manpower from BQ and Huwaila can be deployed to cope with increased workload.

Conclusion

Preserving critical functions & System effective availability of major equipment and during any operational or maintenance intervention by operator/maintenance technician, before proceeding to remote location, requisite resources can be proactively arranged which resulted in saving time and quality of intervention (Maintenance / Operations corrective actions) can be executed to avoid recurrences and achieve uninterrupted production. Successful concept deployment resulted in achieving multiple benefits for remote facilities operation and can be deployed in ensuring effective & optimized plant operations.

Acknowledgement

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Glossary

MPP	– Multiphase pump
MPFM	– Multiphase fluid metering
FMEA	– Failure mode & Effect Analysis
BOPD	– Barrels Oil per day
BWPD	– Barrels water per day
WHCIP	– Well head closed in pressure
TSS	– Total suspended solids
PTS	– Potable Test Separator

MSM – Multi Selector Manifold
KOD – Knock out drum
FDIS – Field Data Integration System
RCA – Root Cause Analysis
DCS – Distributed Control System
RDS – Remote Degassing Station
CDS – Central Degassing Station
F&G – Fire & Gas
VMS – Visual Management Board
PLC – Programmable Logic Controller
KOD – Knockout Drum
DGS – Downgraded Situation
RTU – Remote Telemetry Units
ACM – Advanced Communication modules
ICS – Instrumented control system

References

Remote stations Operating Manual, DCS and PI system data